CoreOps Week-2: Assignments

Cluster Build: This is the base environment on which *all experiments* will run.

Provision Nodes: Launch **two Ubuntu EC2 instances**:

Node	**Purpose**	**Recommended Size**
control-plane	Kubernetes API + scheduler + controller-manager + etcd	t3.medium
worker-node	Workload execution	t3.small

Security Group:

- 6443/tcp → Control Plane API
- 22/tcp → SSH
- All traffic between nodes allowed

Install container runtime (containerd)

Run on BOTH nodes:

- sudo apt update -y
- sudo apt install -y containerd
- sudo mkdir -p /etc/containerd
- containerd config default | sudo tee /etc/containerd/config.toml
- sudo systemctl restart containerd
- sudo systemctl enable containerd

Disable swap

- sudo swapoff -a
- sudo sed -i '/ swap / s/^/#/' /etc/fstab

Install kubelet, kubeadm, kubectl

```
sudo apt update
sudo apt install -y apt-transport-https ca-certificates curl

curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.29/deb/Release.key \
  | sudo gpg --dearmor -o /etc/apt/trusted.gpg.d/kubernetes-apt-keyring.gpg

echo "deb https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /" \
  | sudo tee /etc/apt/sources.list.d/kubernetes.list

sudo apt update
sudo apt install -y kubeadm kubelet kubectl
sudo apt-mark hold kubeadm kubelet kubectl
```

Initialize control plane

Run on Control Plane: sudo kubeadm init --pod-network-cidr=10.244.0.0/16

Configure kubectl:

- mkdir -p $HOME/.kube
- sudo cp /etc/kubernetes/admin.conf $HOME/.kube/config
- sudo chown $USER:$USER $HOME/.kube/config

Install Flannel CNI

kubectl apply -f
https://github.com/flannel-io/flannel/releases/latest/download/kube-flannel.yml

Join worker node

Use the JOIN command given by kubeadm.

Verify: kubectl get nodes -o wide

CLUSTER READY

Now begin **guided assignments for each CoreOps topic**.